OPTION B / 1B | ROUND 1

IEEE-CIS Fraud Detection

Project introduction and course submission guide

Release	Due	Choice
November 5, 2026	November 26, 2026	Complete Option A or Option B

Project background

Fraud screening needs to identify suspicious purchases while avoiding unnecessary declines of legitimate payments. This project compares machine learning approaches on a large collection of real e-commerce transactions.

The data contains transaction, product, device, and identity-related attributes. Participants can engineer additional features to improve fraud predictions. Better detection can reduce fraud losses and the disruption caused by false alerts. For this course, particular attention is given to how models perform on later transactions as behavior changes over time.

Prediction task

Estimate whether an anonymous online payment is fraudulent. Each sample is one transaction, identified by **TransactionID**. Predict **P(isFraud = 1)**.

590,540 labeled transactions; approximately 3.5% fraud, according to the course briefing.

Evaluation

The evaluation metric is **ROC-AUC**, computed from predicted probabilities and observed binary labels. Higher is better. The course evaluates this task using a fixed scorer and hidden test cases.

Use continuous risk scores or probabilities. Changing a classification threshold alone does not change the ROC-AUC of the same scores.

Course assessment

50 points: model performance. **50 points:** human intellectual contribution.

Data and final submission

Files in data/

File	Purpose
train_transaction.csv	Training transactions, including isFraud labels.
train_identity.csv	Identity information for a subset of training transactions.

Join the two training tables using TransactionID. Some transactions have no corresponding identity record. Keep these transactions in the modeling population and check the effect of missing identity information.

Complete a reproducible study

Understand and preprocess the data; choose an appropriate split; establish a simple CPU baseline; test feature or model improvements; validate using ablations and error analysis; and explain how to reproduce the main results. Save configurations, experiment outputs, and reasoning as the work proceeds.

What you must submit

Submit one ZIP file for your chosen option. The ZIP must contain **(1) your notebook and full workspace**, including code, configurations, data-processing steps, and experiment logs; **(2) the complete coding-agent session directory or directories** for the project, with the information needed to locate the relevant interactions; and **(3) Evidence_Card.md**, documenting **at least four key interaction moments** that demonstrate your intellectual contribution. Include the actual session files, not only a path on your computer. Evidence cards provide an index to the full records and do not replace them.

Accountability

You remain responsible for the code, data split, metric, and conclusions. Attribute ideas honestly and preserve real interactions and experiment results. A well-explained result without a stable gain can still demonstrate contribution. Fabricated conversations, misrepresented authorship, and fabricated experiments constitute cheating; cheating receives zero.

Evidence cards

Create Evidence_Card.md with at least four distinct moments. For each moment, use these four fields:

Category. Choose the relevant category A-F below.

Conversation evidence. Locate the original interaction using a screenshot, turn link, or timestamp. Identify the submitted session file so the TA can find it.

My contribution. State what you personally observed, proposed, checked, interpreted, or decided, and explain your reasoning.

Impact. Describe what changed in the experiment, pipeline, decision, or understanding. Use actual results and ROC-AUC where relevant.

Evidence categories

A Problem Framing / Reframing; **B** Hypothesis Generation
C Evaluation / Challenge / Correction; **D** Experiment Feedback / Iteration
E Decision / Trade-off; **F** Knowledge Update / Insight

At least four moments are required; you do not need one card for every category. Distinguish your contribution from the agent's suggestions and submit the actual sessions from your own project.

Human intellectual contribution: 50 points

Dimension	What your evidence should show
Problem Analysis & Insight	Your task-specific observations, analysis, or hypotheses, with reasons.
Decision-Making	Justified choices about research order, alternatives, and resources.
Critical Verification	Specific concerns, meaningful checks, and interpretation of the results.
Understanding	An explanation of what the agent did and why key actions affect the result.

Each dimension accounts for 25% of the contribution component: 12.5 points out of the 100-point project total.

Before submitting

Check that the notebook opens, the workspace identifies how to reproduce the main results, every claimed result has supporting records, all project sessions are included, and each evidence card points to a real interaction.